Credit Risk Analysis Dashboard

994
Total Loans

552
Total Defaults

0.56
Default Rate %

loan_grade

All

age_group

- [ ] < 25
- [ ] 25-34

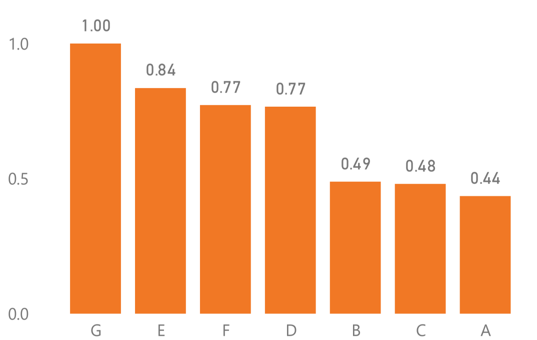
loan_intent

- [ ] Debtconsolidation
- [ ] Education
- [ ] Homeimprovement

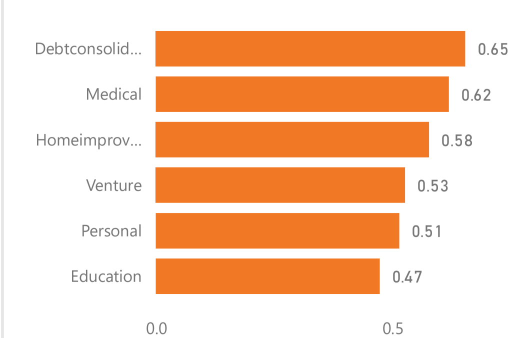
person_home_ownership

- [ ] MORTGAGE
- [ ] OTHER
- [ ] OWN
- [ ] RENT

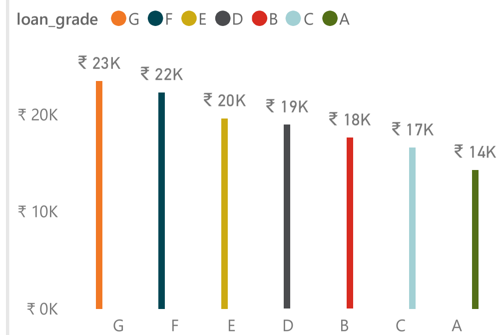
Default Rate by loan grade



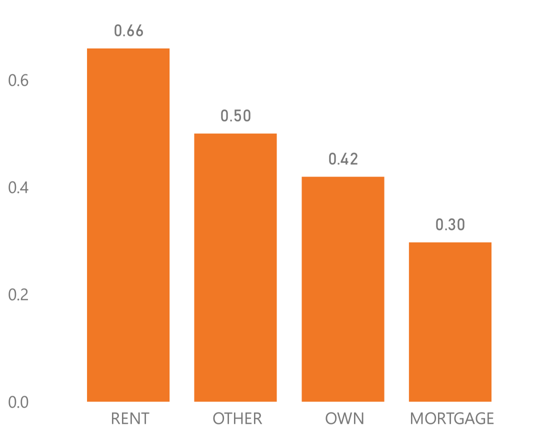
Default Rate by loan intent



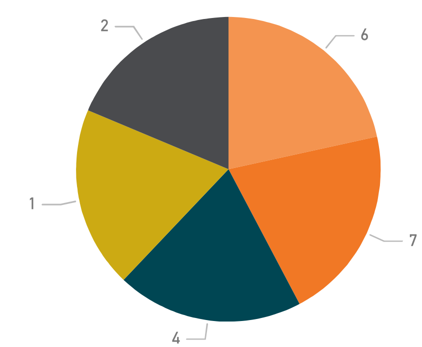
Average Loan Amount by loan grade



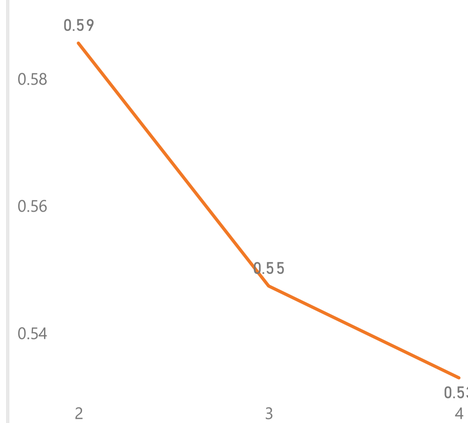
Default Rate by home ownership



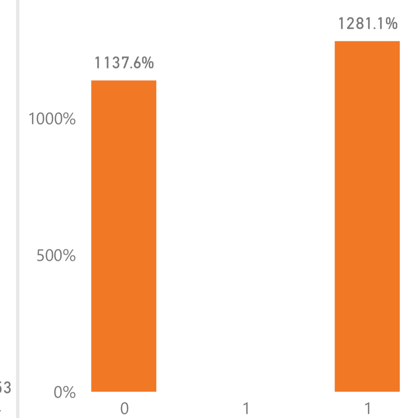
Default Rate vs emp length



credit history lenghth vs default rate



Average Interest Rate by loan status



Executive Summary

- This analysis of the Credit Risk dataset integrates SQL, Excel and Power BI to explore 30 000 + customer loan records.
- After cleaning and transformation, the dashboard identifies key drivers of loan default and risk profiles across borrower segments.
- The overall default rate is ≈ 9 %. Defaults are most common among lower loan grades (E & F), renters, and borrowers with short employment and credit histories.
- Higher interest rates and lower incomes also correlate with greater risk.

Key Insights

- Overall default rate ≈ 9 %; Grades E & F exceed 15 %.
- Renters and low-income borrowers show the highest risk.
- Default likelihood drops as employment and credit history lengthen.
- Personal and Medical loan intents have above-average defaults.
- Higher interest rates correlate with higher defaults.

Recommendations

1. Focus growth on Grade A–C loans and lower-risk segments.
2. Tighten underwriting for high-interest and short-tenure borrowers.
3. Introduce financial literacy & repayment support for renters and low-income customers.
4. Incorporate credit-history length and employment stability in risk scoring models.
5. Monitor portfolio mix monthly to maintain target default rate < 8 %.

Prepare by Tran Thi Minh Thu| Tools: SQL · Excel · Power BI | Data Source: Kaggle Credit Risk Dataset